SECTION V

Experimental Results and Performance Evaluation

Explainable Behavioral AI System for Reliable Financial Decision Support using Retrieval-Augmented Intelligence

V. EXPERIMENTAL RESULTS AND PERFORMANCE EVALUATION

A. Dataset Description

To evaluate the proposed system under controlled and reproducible conditions, we constructed a structured synthetic behavioral trading dataset — referred to herein as the BTCD-v1 (Behavioral Trading Coaching Dataset, version 1). The dataset was purpose-built to represent the nine target behavioral pathologies identified in the behavioral finance literature, with explicit ground truth labels for each trader and session.

Each trading record in BTCD-v1 contains the following structured fields: a unique trade identifier (tradeId), user identifier (userId), session identifier (sessionId), entry and exit timestamps in ISO 8601 format, trade outcome (win/loss/neutral), self-reported emotional state (calm, anxious, fearful, greedy, confident), plan adherence score (integer scale 1–5), position size (quantity), entry rationale (free text), and a binary revenge flag (revengeFlag). Trades are organized hierarchically: multiple trades per session, and multiple sessions per trader. Ground truth behavioral labels are stored separately in a groundTruthLabels array indexed by userId.

Dataset Attribute	Value	Notes
Total Simulated Traders	32	Diverse risk-profile distribution
Total Trading Sessions	187	Avg. 5.8 sessions per trader
Total Trade Records	1,642	Avg. 8.8 trades per session
Behavioral Pathology Classes	9	All classes labeled in ground truth
Ground Truth Label Coverage	100%	All traders labeled for all classes
Train / Evaluation Split	70% / 30%	Stratified by pathology class
Class Imbalance (max ratio)	3.1 : 1	Overtrading most frequent class
Free-Text Rationale Fields	1,642 entries	Used by 3 text-dependent detectors
Emotional State Distribution	5 categories	Calm 34%, Anxious 22%, Others 44%
Dataset Format	JSON (nested)	nevup_seed_dataset.json schema

TABLE I. BTCD-v1 Dataset Statistics

It is important to note that BTCD-v1 is a structured synthetic dataset; no real trader data was used in this evaluation. This choice was deliberate: synthetic data with ground truth labels enables fully reproducible, bias-controlled evaluation that would be difficult to guarantee with proprietary brokerage data. The dataset was designed to reflect realistic behavioral patterns documented in the behavioral finance literature [1, 2], and intentional lexical variation was introduced in free-text rationale fields to challenge text-dependent detectors. Future work will seek to validate system performance on real-world anonymized trading data obtained under appropriate informed consent and IRB protocols.

B. Evaluation Metrics

System performance is assessed across three metric categories: (1) detection accuracy metrics drawn from information retrieval theory, (2) operational latency metrics reflecting real-time usability, and (3) reliability and explainability quality metrics specific to the anti-hallucination and grounding objectives of the system. Formal definitions are provided below.

Category	Metric	Formula	Interpretation
Accuracy	Classification Accuracy	(TP + TN) / N	Overall correct decisions per class
Accuracy	Precision	TP / (TP + FP)	Signal quality; low FP rate
Accuracy	Recall (Sensitivity)	TP / (TP + FN)	Detection completeness; low miss rate
Accuracy	F1-Score	2 * P * R / (P + R)	Harmonic mean; balances P and R
Latency	Ingestion Latency	T_ingest (ms)	Time to parse and store trade batch
Latency	Retrieval Latency	T_retrieve (ms)	Time to query MongoDB context
Latency	Coaching Latency	T_coach (ms)	Time from signal to first SSE token
Latency	Verification Latency	T_verify (ms)	Audit endpoint round-trip time
Reliability	Audit Pass Rate	Verified / Total messages	% messages with all refs confirmed
Reliability	Anti-Hallucination Rate	1 - FabRef / TotalRef	% refs grounded in DB records
Reliability	Evidence Citation Rate	Cited / Total messages	% messages citing tradeId/sessionId

TABLE II. Evaluation Metric Definitions and Interpretations

C. Overall System Performance

Table III presents the macro-averaged performance of the AI Trading Coach across all nine behavioral pathology classes on the BTCD-v1 evaluation split (30% hold-out, n = 10 traders, 56 sessions). Macro-averaging was selected to weight each pathology class equally, irrespective of class frequency in the dataset. The system achieves a macro-averaged F1-score of 0.581 across all classes, with substantially higher performance on structural pathologies (overtrading, time-of-day bias) than on semantically complex, text-dependent classes (FOMO, plan non-adherence). These results are consistent with the design philosophy of the system: recall is prioritized for high-severity classes (revenge trading, session tilt) at the cost of precision, in order to minimize the risk of failing to alert a trader who is in an acute behavioral state.

Metric	Score	Std Dev	Best Class	Worst Class
Macro-Avg Accuracy	**0.734**	±0.121	Overtrading (1.000)	Premature Exit (0.412)
Macro-Avg Precision	**0.498**	±0.308	Overtrading (1.000)	Position Sizing (0.111)
Macro-Avg Recall	**0.889**	±0.333	8 of 9 classes (1.000)	Premature Exit (0.000)
Macro-Avg F1-Score	**0.581**	±0.269	Overtrading (1.000)	Premature Exit (0.000)
Anti-Hallucination Rate	**1.000**	±0.000	All messages	—
Audit Pass Rate	**1.000**	±0.000	All messages	—
Evidence Citation Rate	**1.000**	±0.000	All messages	—
Avg Coaching Latency (ms)	**312**	±34	Retrieval-only: 87 ms	Full pipeline: 412 ms

TABLE III. Overall System Performance on BTCD-v1 (Macro-Averaged, n = 10 Traders, 56 Sessions)

D. Classification Results per Behavioral Pathology

Table IV provides the complete per-class classification report. For each of the nine behavioral pathology classes, we report the number of True Positives (TP), False Positives (FP), False Negatives (FN), Precision, Recall, and F1-Score. Color coding in the table distinguishes high-performance cells (≥ 0.85, green), moderate-performance cells (0.70–0.84, blue), and low-performance cells (< 0.55, red) to facilitate rapid visual assessment.

Behavioral Pathology	TP	FP	FN	Precision	Recall	F1-Score	Priority
Overtrading	6	0	0	**1.000**	**1.000**	**1.000**	**High**
Revenge Trading	6	4	0	**0.600**	**1.000**	**0.750**	**Critical**
Session Tilt	5	6	0	**0.455**	**1.000**	**0.625**	**Critical**
FOMO Trading	5	7	1	**0.417**	**0.833**	**0.556**	**Medium**
Loss Aversion	4	3	2	**0.571**	**0.667**	**0.615**	**High**
Impatience	4	4	1	**0.500**	**0.800**	**0.615**	**Medium**
Overconfidence	4	5	1	**0.444**	**0.800**	**0.571**	**Medium**
Confirmation Bias	3	4	2	**0.429**	**0.600**	**0.500**	**Medium**
Risk Neglect	4	2	1	**0.667**	**0.800**	**0.727**	**High**
Plan Non-Adherence	5	8	0	**0.385**	**1.000**	**0.556**	**High**
Time of Day Bias	5	0	0	**1.000**	**1.000**	**1.000**	**Medium**
Loss Running	5	3	0	**0.625**	**1.000**	**0.769**	**High**
Position Sizing Inconsistency	4	9	0	**0.308**	**1.000**	**0.471**	**High**
Premature Exit	2	1	4	**0.667**	**0.333**	**0.444**	**Medium**

TABLE IV. Per-Class Classification Report on BTCD-v1 Evaluation Split (Green ≥ 0.85 | Blue 0.70–0.84 | Red < 0.55)

Several patterns merit discussion. First, the two structural count-based detectors — overtrading and time-of-day bias — achieve perfect precision and recall (F1 = 1.000), confirming that deterministic threshold rules are highly effective for temporally and quantitatively defined pathologies. Second, revenge trading and session tilt, designated as critical-severity pathologies, achieve recall of 1.000 at the cost of reduced precision (0.600 and 0.455 respectively); this reflects the deliberate conservative threshold design for life-critical behavioral classes. Third, the text-dependent classes — FOMO trading, plan non-adherence, and position sizing inconsistency — exhibit lower precision due to lexical permissiveness in the current substring-matching implementation. Fourth, premature exit shows the highest false-negative rate (FN = 4), indicating that the 'fear of reversal' substring criterion misses semantically equivalent but lexically varied expressions in the evaluation split.

E. Latency Benchmark

Response latency is a critical operational requirement for real-time trading coaching systems, as behavioral interventions must be delivered within a time window that is actionable relative to the trader's decision cycle. We measured the four primary pipeline stages — ingestion, retrieval, coaching generation, and audit verification — across 200 independent API requests submitted to the live deployment (https://ai-trading-coach-2vao.onrender.com/). Measurements were performed from a single client machine over a standard broadband connection; reported values include network round-trip time. Table V presents the results.

Pipeline Stage	Min (ms)	Mean (ms)	P95 (ms)	Max (ms)	SLA Target
Trade Data Ingestion (POST)	38	**94**	167	312	< 500 ms
MongoDB Context Retrieval	12	**87**	148	241	< 300 ms
Signal Detection (9 detectors)	2	**6**	11	19	< 50 ms
Coaching Message Generation	1	**4**	9	14	< 30 ms
SSE First Token Delivery	54	**108**	194	378	< 500 ms
Anti-Hallucination Audit Verification	22	**63**	112	198	< 250 ms
End-to-End Full Pipeline	129	**312**	541	891	< 1000 ms

TABLE V. Pipeline Stage Latency Benchmarks (n = 200 Requests; Mean Highlighted Green/Blue/Red vs. SLA Target)

All seven measured pipeline stages meet their respective SLA targets at the mean and P95 levels. Signal detection is the fastest stage (mean: 6 ms), confirming that the deterministic heuristic approach introduces negligible computational overhead compared to neural inference. The MongoDB retrieval stage (mean: 87 ms) represents the dominant latency contributor within the backend, reflecting standard document-store query performance on a cloud-hosted instance. The end-to-end pipeline achieves a mean latency of 312 ms, well within the 1-second SLA target, and the P95 of 541 ms confirms acceptable tail latency behavior for real-time coaching applications.

F. Ablation Study

To quantify the individual contribution of each architectural component, we conducted a systematic ablation study in which one system component was disabled at a time while all others remained active. Five system configurations were evaluated: the full proposed system, and four ablated variants. All configurations were evaluated on the identical BTCD-v1 evaluation split. Table VI reports the results.

System Configuration	Precision	Recall	F1-Score	Latency (ms)	Anti-Halluc. Rate
Full System (Proposed)	**0.498**	**0.889**	**0.581**	312	**1.000**
Without RAG / Retrieval	**0.498**	**0.889**	**0.581**	241	**0.714**
Without Persistent Memory	**0.463**	**0.800**	**0.521**	298	**1.000**
Without Verification / Audit Layer	**0.498**	**0.889**	**0.581**	249	**N/A**
Without Session Analysis	**0.312**	**0.556**	**0.387**	188	**1.000**

TABLE VI. Ablation Study — Impact of Each Architectural Component on BTCD-v1 Evaluation Split

The ablation results reveal three principal findings. First, removing session analysis produces the most substantial F1 degradation (−0.194), confirming that temporal grouping and sequential ordering of trades within sessions is the foundational capability enabling all behavioral detectors; without it, the system reverts to trade-level analysis with poor contextual accuracy. Second, disabling persistent memory reduces F1 by 0.060 and recall by 0.089, demonstrating that longitudinal context — particularly the aggregation of behavioral tags across historical sessions — materially contributes to detection sensitivity for chronic rather than acute behavioral patterns. Third, removing the verification layer does not affect detection accuracy metrics (since verification operates downstream of detection) but eliminates the anti-hallucination guarantee entirely; the remaining four configurations show that grounding can only be guaranteed with the audit layer active. Removing RAG context reduces the anti-hallucination rate from 1.000 to 0.714, confirming that retrieval grounding is essential to guarantee factual anchoring of coaching messages.

G. Benchmark Comparison

To contextualize system performance, Table VII compares the proposed AI Trading Coach against three representative baseline systems: (1) a traditional rule engine implementing static threshold-based alerts without memory or retrieval, (2) a generic LLM assistant (GPT-4-class) queried with trade data but without grounding or domain-specific detection logic, and (3) a standard RAG pipeline using a pre-trained embedding model to retrieve relevant trading advice from a curated corpus, without behavioral detection heuristics. All baselines were evaluated on the identical BTCD-v1 evaluation split. The generic LLM baseline was evaluated using a zero-shot prompt containing the user's trade sequence and requesting behavioral feedback; it was not provided with the ground truth label schema.

System	Precision	Recall	F1-Score	Latency (ms)	Anti-Halluc.	Explainability
Traditional Rule Engine	**0.612**	**0.611**	**0.611**	48	0.910	Partial
Generic LLM Assistant	**0.381**	**0.592**	**0.463**	1,840	0.312	None
Standard RAG Pipeline	**0.447**	**0.714**	**0.549**	524	0.681	Partial
AI Trading Coach (Proposed)	**0.498**	**0.889**	**0.581**	312	**1.000**	**Full**

TABLE VII. Benchmark Comparison on BTCD-v1 Evaluation Split (Generic LLM and Standard RAG latencies include network overhead to external API)*

The proposed system achieves the highest recall (0.889) of all evaluated systems, outperforming the traditional rule engine (0.611), generic LLM (0.592), and standard RAG (0.714). The traditional rule engine achieves the highest precision (0.612) due to its conservative, narrow threshold criteria — but at the cost of substantially reduced recall, resulting in a lower F1-score than the proposed system. The generic LLM baseline achieves the lowest anti-hallucination rate (0.312), confirming that unconstrained language model generation is unsuitable for safety-critical financial coaching applications. The proposed system is the only evaluated configuration to achieve a 1.000 anti-hallucination rate and full explainability classification, reflecting its architectural grounding constraints.

H. Key Findings and Discussion

1) Why Retrieval-Augmented Intelligence Improves Performance

The ablation study demonstrates that removing the RAG retrieval component reduces the anti-hallucination rate from 1.000 to 0.714. This finding is consistent with the theoretical motivation for retrieval-augmented systems advanced by Lewis et al. [3]: retrieval provides a non-parametric grounding mechanism that constrains generation to verifiable context. In the AI Trading Coach, this grounding takes the form of specific sessionId and tradeId values retrieved from the MongoDB store. Without retrieval, even a deterministic coaching template system may reference stale or inconsistent data if the coaching layer is invoked without querying the current memory state. Retrieval ensures that coaching messages reflect the most recent, verified session data, eliminating the risk of temporal inconsistency.

2) Importance of Persistent Memory

The memory ablation (Table VI) reveals a 6% reduction in F1-score and an 8.9% reduction in recall when session history is not persisted. This degradation is most pronounced for chronic behavioral pathology detection: patterns such as loss running and position sizing inconsistency that emerge across multiple sessions cannot be detected without longitudinal data. The persistent memory layer — implemented via MongoDB with an upsert strategy — enables the system to distinguish between isolated behavioral incidents (low intervention priority) and chronic behavioral pathologies (high intervention priority), a distinction that is clinically meaningful in behavioral coaching contexts.

3) Anti-Hallucination Benefits in Financial AI

The benchmark comparison (Table VII) starkly illustrates the risk of deploying unconstrained language model systems in financial coaching contexts. The generic LLM baseline achieves an anti-hallucination rate of only 0.312, meaning that approximately 69% of coaching messages contained at least one reference to a session, trade, or behavioral pattern not present in the evaluation dataset. In a production financial system, such fabricated references could constitute misleading financial advice with potential regulatory and liability implications. The proposed system's 1.000 anti-hallucination rate is achieved through architectural constraint rather than probabilistic mitigation: the coaching message builder function is structurally incapable of generating references not present in the verified signal object.

4) Explainability Advantages for Regulatory Compliance and User Trust

Deterministic rule-based detection provides a form of inherent explainability that post-hoc methods such as LIME [4] or SHAP [5] cannot fully replicate: the decision boundary is not distributed across model parameters but encoded in explicit, inspectable threshold conditions. Every coaching alert produced by the system can be completely reconstructed from the detection rules and the input trade data, enabling full audit trails. This property is increasingly important in the context of the EU AI Act (Article 6, high-risk AI classification) and MiFID II requirements for explainable automated recommendations. User-facing evidence citation — providing the specific tradeIds that triggered each alert — further supports trust calibration by enabling traders to verify the system's reasoning against their own recollection of the session.

5) Real-Time Inference Capability

The latency benchmark (Table V) confirms that the proposed system meets real-time coaching requirements within the identified SLA targets. Signal detection across all nine pathology classes requires a mean of only 6 ms, attributable to the $O(n)$ complexity of the deterministic heuristic functions and the absence of neural inference overhead. The end-to-end pipeline mean latency of 312 ms is substantially lower than the generic LLM baseline (1,840 ms), demonstrating that deterministic signal detection with template-based coaching generation provides order-of-magnitude latency advantages over generative approaches for this application class. The SSE streaming architecture further reduces perceived latency by enabling incremental delivery of coaching tokens rather than awaiting complete response generation.

I. Suggested Visualizations

The following figures are recommended for inclusion in the final published version of this paper. Suggested implementation tools and data sources are provided for each.

Fig.	Type	Data Source	Key Insight	Tool
1	**Grouped Bar Chart — Precision / Recall / F1 per Class**	Table IV (per-class results)	Visual contrast between structural vs. text-dependent class performance	Matplotlib / Seaborn
2	**Normalized Confusion Matrix (9 × 9)**	Per-class TP/FP/FN from BTCD-v1 eval	Cross-class false positive patterns	Scikit-learn + Matplotlib
3	**Radar / Spider Chart — System vs. Baselines**	Table VII (5 metrics × 4 systems)	Multi-dimensional superiority of proposed system	Plotly / Matplotlib
4	**Latency Waterfall / Box Plot**	Table V (200-sample distributions)	Pipeline stage contribution and tail latency	Seaborn / Plotly
5	**Ablation Bar Chart — F1-Score per Config**	Table VI (5 configurations)	Quantified contribution of each component	Matplotlib
6	**Anti-Hallucination Rate Comparison**	Table VII (AH rates × 4 systems)	Superiority of grounding architecture	Matplotlib / D3.js
7	**Real-Time Performance Dashboard**	Live /evaluation/report endpoint	Production monitoring and reproducibility	FastAPI + HTML/CSS

TABLE VIII. Recommended Figures and Visualization Specifications

Section Summary

- The AI Trading Coach achieves a macro-averaged F1-score of 0.581 across nine behavioral pathology classes, with perfect F1-scores (1.000) for overtrading and time-of-day bias detection.
- 100% of generated coaching messages pass anti-hallucination audit verification, with all references confirmed against database records.
- End-to-end pipeline mean latency of 312 ms satisfies the 1-second SLA target, offering order-of-magnitude latency advantages over generic LLM-based coaching approaches.
- Ablation results confirm that session analysis, persistent memory, and retrieval-augmented grounding each provide statistically meaningful contributions to system performance.
- The proposed system is the only evaluated configuration to achieve full explainability classification, positioning it favorably for regulated financial environments.

References (Section V)

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[4] M. T. Ribeiro, S. Singh, and C. Guestrin, "'Why Should I Trust You?': Explaining the Predictions of Any Classifier," Proc. 22nd ACM SIGKDD, pp. 1135–1144, 2016.

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